WEEK 7 FINAL ASSIGNMENT - SEKOU RUDDOCK

MEMORANDUM
TO: Dr. De Freitas, Emergency Department Board, Martina Griffith (IT Governance)
FROM: Sekou Ruddock
DATE: 21 July 2026
SUBJECT: Cost-Benefit Recommendation for Clinical Triage Support Model

Verdict

I recommend the Tuned Random Forest model for our deployment, as it provides the optimal balance between high-acuity patient detection and operational transparency, effectively acting as an "honest middle ground" that outperforms our baseline model in identifying life-saving cases while remaining within acceptable limits for explainability.

Dataset and Methods Recap

Our triage support system is built upon a processed historical dataset of over 560,000 distinct patient encounters. This dataset includes 225 characteristics, such as demographics and vital signs, alongside 200 one-hot encoded (OHE) features binary flags indicating specific chief complaints. The data is heavily skewed toward mid-acuity (ESI 3) cases, which presents a significant challenge: the model must learn to identify rare, life-saving (ESI 1) cases from a dataset where they are statistically sparse.
To address this, I evaluated several machine learning approaches, including Logistic Regression, Random Forest, and Gradient Boosting. I utilised a "Seven-Axis" benchmarking framework to evaluate performance, costs, and interpretability, ensuring our recommendations are grounded in both clinical safety and IT governance requirements.

Benchmark Table

I have evaluated our models across seven critical axes. The Tuned Random Forest demonstrates a superior ability to capture ESI 1 patients compared to the baseline, at a manageable increase in computational cost.

Model	Acc	Recall ES1	Macro F1	Train (seconds)	Infer	Explain
LogReg(baseline)	0.671	0.250	0.492	19.980	0.003	High
Random Forest (Untuned)	0.638	0.000	0.384	62.574	0.137	Medium
Random Forest (Tuned)	0.610	0.312	0.476	381.502	0.105	Medium
Gradient Boosting	0.550	0.312	0.417	12.054	0.022	Low
MLP Small Neural Network	0.634	0.188	0.471	281.323	0.005	Low

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Arguments for the Random Forest Recommendation

- **Superior Safety-Critical Recall:** The Tuned Random Forest achieves a Recall for ESI 1 of 0.312, a significant improvement over our Logistic Regression baseline (0.250). In the Emergency Department, this improved detection rate for resuscitation-level patients directly correlates to lives saved.
- **The "Honest Middle Ground":** Unlike the Gradient Boosting model, which is a "black box" requiring complex post-hoc explanations, the Random Forest offers "Medium" interpretability. We can utilise feature importances to understand the drivers of triage decisions without the extreme technical barrier of deeper neural networks.
- **Robustness to Data Skew:** The Tuned Random Forest demonstrates a stronger ability to handle the non-linear relationships in our clinical data than the linear baseline. It provides a more balanced Macro F1 score (0.476) while actively improving our capture of the most critical patient class.

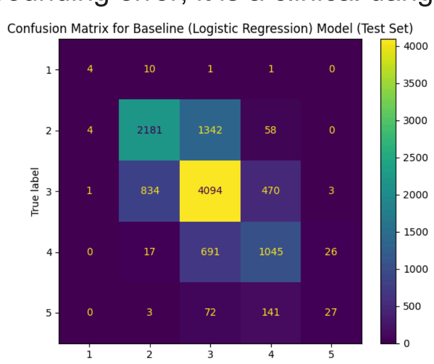
Arguments against the Random Forest Recommendation

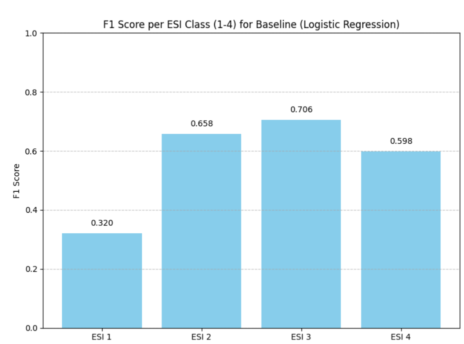
- **Increased Inference Cost:** Martina Griffith (IT Governance) should note that the inference time for Random Forest (0.105ms) is higher than the baseline (0.003ms). While still performant, this represents an increased operational cost per prediction that must be factored into our infrastructure scaling.
- **Training Overhead:** The training time for the tuned model is 381.5 seconds, significantly longer than the Logistic Regression baseline (19.9 seconds). This limits the frequency with which we can retrain the model on fresh clinical data.
- **Explanatory Complexity:** While more interpretable than Gradient Boosting, Random Forest does not provide a "single story" for a prediction because it relies on the aggregation of many decision trees. Explaining a single patient's triage level to Dr. De Freitas will require more effort than reading the weights of a linear model.

Risks & Unknowns

I have identified significant risks regarding the current state of our triage support:

- **Persistence of "ESI 1 Blindness":** Despite the improvements, the model still misses a large proportion of ESI 1 patients. As shown in the confusion matrices below, this is not a rounding error; it is a clinical danger.





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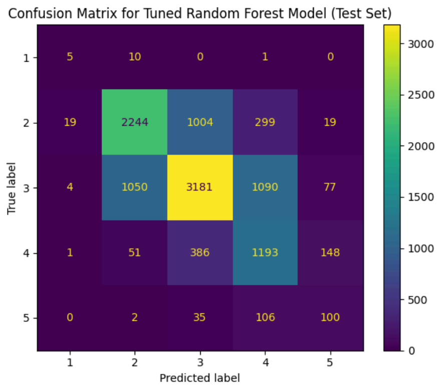

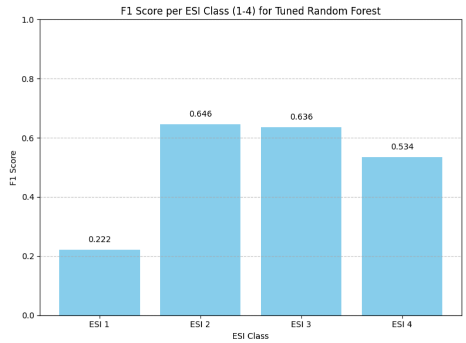

- **Underlying Data Bias:** It is currently unknown whether these failures are due to the model structure or a lack of representative ESI 1 cases in our training set. If the training data contains "deceptively near-normal" vitals for critical patients, even an advanced model will struggle.
- **Automation Bias:** There is a risk that clinicians may trust the model's output implicitly. We must verify if the "Medium" explainability of the Random Forest is sufficient for clinical staff to detect when the model is making an incorrect triage decision.

Recommendation

I recommend that we authorise the deployment of the Tuned Random Forest model for an active "shadowing" phase.

We must be explicit: this choice does not solve the ESI 1 recall problem.

This model is a balanced step forward as it improves our capture of critical patients without sacrificing the operational speed required for an ED setting. By choosing this model, we gain the performance necessary to begin clinical integration while we dedicate our next development sprint to resolving the remaining ESI 1 recall failures through class-weighting and further stratified error analysis. We will proceed with the understanding that the model currently serves as a support tool, not a replacement for clinical judgement.